One Blendshape Track, Any Avatar Rig: Real-Time Speech-Driven Faces in the Browser

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Measured 2026-09-22 to 2026-11-04 on production traffic

98.1% of user-uploaded rigs with morph targets animate from one Audio2Face-3D track, with no per-rig retargeting step, at a median **3.8 s** from text to speaking face.

1 Problem

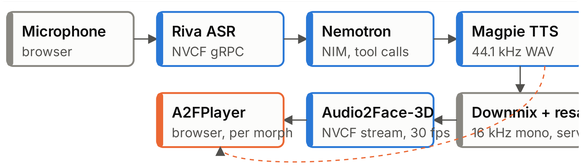
Speech-driven facial animation ships as a **fixed blendshape vocabulary**, but the avatars people actually bring to a browser do not share one.

Audio2Face-3D emits 52 ARKit shapes plus tongue and head pose at 30 fps. On a platform where users upload, fork, auto-rig, and reconstruct their own characters, the receiving mesh exposes ARKit names, VRM vowels (`Aa Ih Ou Ee Oh`), Oculus visemes (`viseme_aa` …), vendor spellings, or no morph targets at all. In our sample, **54.9%** of files carry no facial morphs at all, and **11 of 211** of the morphed files expose no ARKit mouth shape under any spelling.

Goal. Drive every rig a user brings from the same track, in the browser, with no per-rig authoring, no model-side retargeting, and no frozen faces when coverage is partial.

Constraints. Audio must stay perceptually synchronized with the mouth; the client is a web page (no native runtime); rigs are untrusted glTF with meshopt and Draco compression, quantization, and arbitrary node structure.

2 Application of NVIDIA technology



End-to-end path for one spoken turn. Everything above the browser runs as hosted NVIDIA Cloud Functions over gRPC; the browser receives the original audio and a JSON track and does all retargeting locally.

- **Riva ASR** (Canary, multilingual) transcribes the user's microphone audio.
- **Nemotron** (`nemotron-3.5-lightning-30b-a3b` via NIM) produces the reply and tool calls.
- **Magpie TTS Multilingual** renders speech at 44.1 kHz; the server downmixes and linearly resamples a copy to 16 kHz mono for inference while the browser plays the original.
- **Audio2Face-3D** (NVCF bidirectional stream) returns per-frame weights for 55 shapes; the server packs them as `{fps, blendShapeNames, frames[{t, w[55]}]}`.

Code, data and live demo

Source and measurements
Player, composition tables, coverage and latency scripts, raw data behind every figure.

Live demo
Speak or type; drop in your own glTF and watch which path binds each morph.

3 Method: direct and derived mapping

At attach time the player inspects every morphed mesh once and binds each morph target to one of two paths. Per frame it does no string work: the track is sampled by the **audio element's clock** with linear interpolation, so the mouth cannot drift from the sound even when frames drop.

Direct path

A morph name is canonicalized (case, separators, an `ARKit_` prefix stripped) and, if it resolves to one of the 52 ARKit shapes, it is driven by that track column verbatim.

Derived path

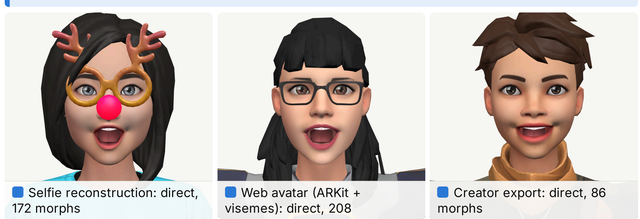
A VRM vowel or Oculus viseme morph v with published ARKit composition c_v receives the weighted mean of the ARKit frame a over its components, the inverse of the authoring map:

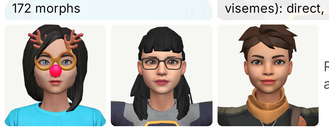
```
w(v) = clamp01( Σk a[k] · cv[k]  /  Σk cv[k] )

// composition tables (excerpt)
viseme_aa: { jawOpen: 0.95 }
viseme_O:  { jawOpen: 0.50, mouthFunnel: 0.55 }
viseme_PP: { mouthClose: 0.90, mouthPressLeft: 0.40, mouthPressRight: 0.40 }
Ou (VRM): { mouthFunnel: 0.70, mouthPucker: 0.40, jawOpen: 0.20 }
```

Suppression rule

A mesh that already has **any direct jaw or mouth shape** gets no derived bindings. Both paths at once double-drive the mouth (Section 6). Meshes with neither path fall back to amplitude lipsync from the playing audio, so a face is never frozen.





Rest poses of the same three rigs (every morph at 0), for comparison.

The same frame on three unrelated rigs. Frame 160 (t = 5.33 s, JawOpen = 0.68) of a production track for the sentence *"Hello, I am a three dimensional agent running in your browser, and my face is animated live by Audio2Face as I speak."*, applied by the shipping player to three avatar files with different authors, tools, and morph vocabularies. No rig-specific code or allowlist exists; new rigs are supported by adding a name composition, never a rig entry.

4 Benchmarking method

- **Rig coverage.** 468 distinct avatar files from human accounts, at most 60 per creation source (seeded, QA and synthetic accounts excluded). Each glTF's JSON chunk is read by HTTP range request and its morph dictionaries are classified by the *shipping* `A2FPlayer.attach()`, not a reimplementation.
- **Latency.** Ten consecutive production calls with the same 22-word sentence; wall clock from request start to the full JSON response (TTS, resample, Audio2Face-3D stream). Rate-limited calls are excluded, not retried.
- **Overshoot.** The real 228-frame track replayed through the composition tables, with and without the suppression rule.

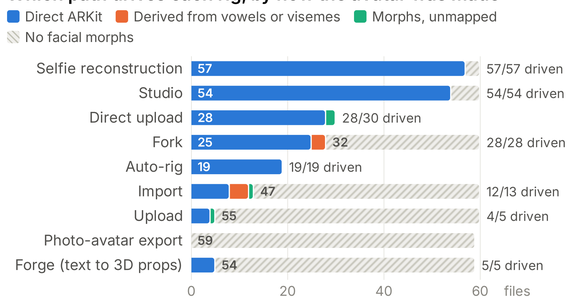
5 Results and outcomes

98.1%
of morphed rigs driven (207 of 211)

3.8 s
median text to animated speech, 7.3 s of audio

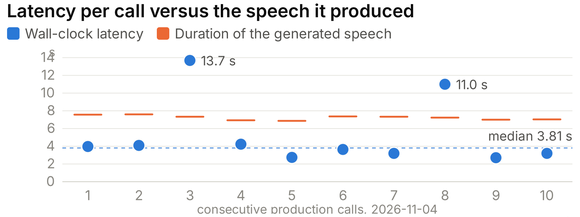
7
rigs rescued only by the derived path (4 still unmapped)

Which path drives each rig, by how the avatar was made



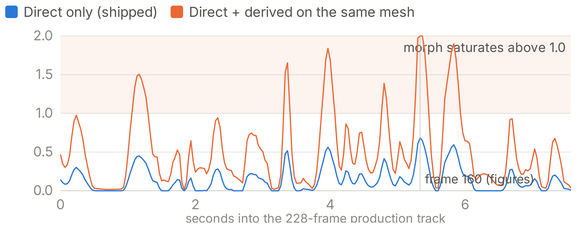
Face-producing sources (selfie, studio, auto-rig) bind by the direct path. The derived path matters for files imported from other ecosystems (Import, Fork): 7 rigs that would otherwise be frozen. Photo-avatar exports and text-to-3D props carry no facial morphs and get amplitude lipsync.

Latency per call versus the speech it produced

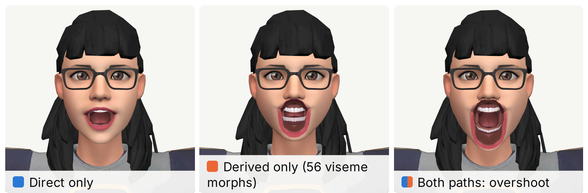


8 of 10 calls finished in 2.7 to 4.3 s for 6.9 to 7.6 s of speech (real-time ratio 0.39 to 0.61); two hit upstream tail latency at 11.0 and 13.7 s. Earlier runs: median 2.96 s (2026-10-08, six calls); 1.29 to 2.19 s with audio supplied directly, skipping TTS (2026-09-22).

6 Ablation: why both paths cannot run at once



Mapping-implied jaw opening over the real track. With the suppression rule off, the ARKit `jawOpen` column and the visemes derived from it stack: the mouth saturates in **23%** of voiced frames (peak 6.1× the authored opening) instead of 0%.



Same rig, same frame. **Derived only** (the rig's ARKit mouth shapes hidden from the player) is what a VRM or Oculus-style rig receives: a usable mouth from visemes alone. **Both** is the failure the rule prevents.

7 Lessons and limitations

- **Name canonicalization beats allowlists.** One normalization function covered vendor spellings (`ARKit_jawOpen`, `jaw_open`, `JawOpen`) that an explicit rig list would have missed.
- **Clock the face from the audio element**, not from `requestAnimationFrame`. Drift was the first bug users noticed and the cheapest to remove.
- **Latency is dominated by upstream tails**, not retargeting (sub-millisecond per frame). Streaming the first frames before TTS completes is the next step.
- **Limits.** Derived weights are a linear inverse of hand-authored compositions; shapes with no vowel analogue (`mouthRollUpper`) are lost on viseme-only rigs. 54.9% of uploaded files have no facial morphs, which no mapping can fix.

References

1. NVIDIA. Audio2Face-3D NIM, blendshape output and gRPC streaming. `docs.nvidia.com/ace/audio2face-3d-microservice`
2. NVIDIA. Riva ASR and Magpie TTS on NVIDIA Cloud Functions. `docs.nvidia.com/nim/riva`
3. Apple. ARFaceAnchor.BlendShapeLocation (52 shapes). `developer.apple.com/documentation/arkit/arfaceanchor/blendshapelocation`
4. VRM Consortium. VRM 1.0 expressions (aa, ih, ou, ee, oh). `github.com/vrm-c/vrm-specification`
5. Meta. Oculus Lipsync viseme reference (15 visemes). `developers.meta.com/horizon/documentation/unity/audio-ovrlipsync-viseme-reference`
6. Khronos. glTF 2.0, KHR_mesh_quantization, EXT_meshopt_compression. `github.com/KhronosGroup/glTF`
7. three.js r0.187.0, GLTFLoader and morph target animation. `threejs.org`